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Descriptor: *Grocery Fashion Retail Consumer Tech Low-Resource Dataset (Lingu10)*

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ABSTRACT The rapid expansion of multilingual applications in e-commerce and customer service operations requires immediate access to conversational datasets that support both low-resource and high-resource languages. The Lingu10-Grocery-FashionRetail-ConsumerTech-LowResource dataset serves as a multilingual domain-specific corpus that researchers and developers can use to advance conversational artificial intelligence (AI) systems. The dataset comprises prompt–response pairs that cover three essential retail sectors: grocery and fashion retail, and consumer technology, across ten languages, representing both popular and minority languages. The dataset comprises English product names accompanied by native language prompts and responses, which preserve authentic cultural and linguistic characteristics. The development of Lingu10 employed script-driven synthetic generation methods to create a resource that provides rich context and diverse content, while preventing repetitive template usage. The dataset release includes complete generation scripts, seed files, and language mappings, which allow users to reproduce results and expand the dataset to new domains and languages. The dataset serves as a valuable resource for advancing multilingual natural language processing through its applications in chatbot training, cross-lingual dialogue systems, low-resource domain adaptation, and culturally aware virtual assistants. The open release of Lingu10 as a dataset aims to speed up research activities between academic institutions and industrial organizations while reducing the performance difference between languages with abundant data and those with limited resources in practical conversational systems.

IEEE SOCIETY/COUNCIL IEEE Computational Intelligence Society

DATA TYPE/LOCATION Conversational text (multilingual, synthetic, domain-specific dialogue corpus); Global (synthetically generated, reproducible via provided scripts).

DATA DOI/PID 10.21227/rb2h-wn79

INDEX TERMS Chatbots, conversational artificial intelligence (AI), human–machine dialogue, multilingual dataset, natural language processing, synthetic dataset.

BACKGROUND

Millions of online customers interact with chatbots as their primary point of contact when they need product information

or support. The systems function adequately for English and Chinese users, but they fail to deliver proper service to users who speak underrepresented languages. The same product or

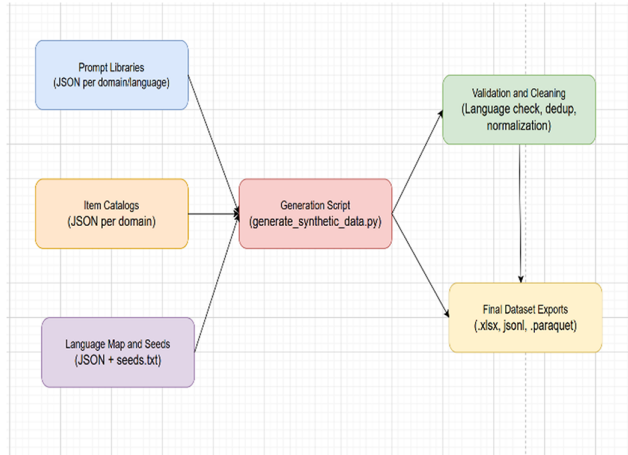


FIG. 1. Pipeline for Lingu10 dataset generation, from inputs (prompts, items, language seeds) through generation and validation to final exports.

service appears well-supported in one language but becomes inaccessible when customers use another language. The existing public conversational datasets were developed to focus on specific, narrow applications [1]. The datasets primarily contain English and several worldwide languages, but their heavy reliance on machine translation results in robotic dialogues that lack cultural understanding. The current systems lack proper coverage of specific domains. The dataset lacks sufficient coverage of retail dialogues about groceries, fashion items, and consumer electronics because these topics represent typical customer service inquiries worldwide.

The Lingu10-Grocery-FashionRetail-ConsumerTech-Low-Resource dataset functions as a solution to this problem. The dataset comprises ten languages, including both popular and less common languages, unified into a single collection for developing retail chatbot systems. The dialogue pairs exist in their original target language without translation and include actual product names to maintain authentic cultural dialogue. The script-based data generation system allows developers to add new domains and languages to the dataset while preventing repetitive template usage. The dataset release aims to support developers and researchers in creating inclusive conversational systems that understand real shopping behaviors and cultural differences.

COLLECTION METHODS AND DESIGN

The Lingu10-Grocery-FashionRetail-ConsumerTech-Low-Resource dataset was generated through a script-based process that maintained balance and cultural accuracy, producing identical results across ten languages in three retail sectors. The process is illustrated in Fig. 1 as a visual representation.

Data Sources and Inputs

The `reproducibility_artifacts/directory` contains three input components, which serve as the foundation for the study.

- 1) The domain prompt libraries (`prompts/*.json`) contain 100 native-language prompt patterns for each domain

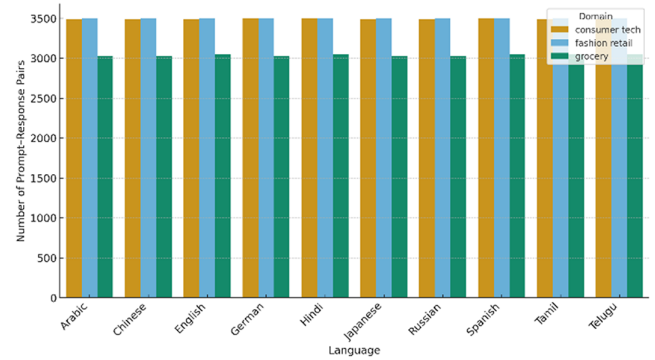


FIG. 2. Distribution of prompt-response pairs in the Lingu10 dataset across ten languages and three retail domains (consumer tech, fashion retail, grocery).

and target language while using placeholders like `{item}` for product insertion.

- 2) The item catalogs (`items/*.json`) contain English product/name lists for each domain, which serve as the basis for creating specific product-based queries.
- 3) The language map and seeds (`languages/languages.json`, `seeds/seeds.txt`) contain ISO language codes and fixed random seeds for Python/NumPy/PyTorch/HF Trainer to generate and sample data deterministically.

Generation Procedure

The script `generate_synthetic_data.py` located in `scripts` generates a balanced dataset that matches each language and domain combination. The script generates pairs of prompts (in the native language) and responses (in the native language) by running through native prompt patterns and selecting items to replace `{item}` placeholders. The system maintains language-specific quotas for achieving balanced representation while it mixes prompt collections to minimize duplicate content.

Language Verification and Post-Processing

The pipeline performs automatic language consistency verification through `langdetect`, which checks both prompts and responses against their designated ISO codes. The system logs any detected mismatches to `language_warnings.txt` for human evaluation. The system performs three operations after processing: it standardizes column headers, removes inconsistent data rows, and eliminates duplicate prompt-response combinations. The system generates output artifacts in three different formats (`.xlsx`, `.jsonl`, and `.parquet`) for subsequent applications. The final dataset contains 100 330 prompt-response pairs across ten languages and three domains. The distribution of pairs is summarized in Fig. 2.

Reproducibility and Extension

The seeds for reproducibility are stored in `reproducibility_artifacts/seeds/seeds.txt`. The batch script (`batch-commands.txt`) enables users to produce the complete dataset through deterministic regeneration [3]. The addition of new

domains and languages requires updating the prompt JSONs, item catalogues, and language map; however, the script will generate balanced outputs with a matching schema.

VALIDATION AND QUALITY

Completeness and Balance

The Lingu10 dataset developers implemented balance as a fundamental principle during its development. The final version of the dataset comprises 100 330 prompt–response pairs, evenly distributed with approximately 10 000 entries for each language. The three domains maintain balanced representation in the dataset because consumer technology, fashion retail, and grocery each contribute approximately 34 900 pairs, 35 000 pairs, and 30 400 pairs, respectively. The balanced distribution of data in the dataset prevents models from developing language or domain-specific biases [2].

Language Consistency

The langdetect library performed an automatic check to verify that each prompt and response appeared in their designated language. The system compared every entry against the ISO code, which corresponded to its language according to the language map. The system recorded all detected language mismatches in a file named `language_warnings.txt` for future evaluation. The detection system produced near-perfect results but occasionally generated false positives when encountering script overlaps between the Chinese and Japanese writing systems. The observed false-positive rate for CJK languages was approximately 0.4%, based on 43 flagged instances across 10 330 CJK-region entries. All warnings recorded in `language_warnings.txt` were manually reviewed by the authors, with automatic resolution applied only after verification.

Deduplication and Cleaning

The post-generation process involved standardizing all column headers and eliminating improperly formatted data entries. The system performed a duplicate detection process on all prompt–response pairs. The duplicate check across 30 language–domain files revealed 110 duplicate entries, which made up 0.11% of the total 100 000+ entries [6]. The team eliminated these duplicate entries to prevent data inflation and maintain a reliable training dataset. The outcomes of these validation checks are summarized in Table I.

Data Integrity

The item, prompt, and response fields in the dataset contain no missing data, as verified by our process. We assessed the naturalness of entries by analyzing the average text length in various languages. The length of prompts spans between 20 and 60 characters, while responses span between 30 and 70 characters, which follows typical conversational patterns.

TABLE I. Validation Results for the Lingu10 Dataset

Check	Result	Accuracy/Error Rate
Missing values	None detected	0% null rate
Duplicate pairs	110 of 100 330	0.11% duplicates (removed)
Language verification	>99.5% match to ISO code	< 0.5% mismatch rate
Balance across languages	~10k per language	± 3% variance
Balance across domains	30k–35k pairs per domain	± 7% variance

Privacy

The Lingu10 dataset contains synthetic dialogues that lack all personally identifiable information, user-generated content, and sensitive data because it was generated through artificial means [5]. The dataset remains secure for sharing purposes because it contains no real-world conversational data, which typically poses privacy risks.

Reproducibility

The dataset provides reproducibility through fixed seeds located in `seeds/seeds.txt` which enables deterministic regeneration of the data [11]. The included batch script allows researchers to reproduce dataset statistics and expand the data collection to additional languages and domains while maintaining consistency with the original release.

RECORDS AND STORAGE

The Lingu10 dataset is organized into domain-specific directories with one file per language, accompanied by supporting artifacts that enable reproducibility and extension. All primary dialogue files are stored in Excel format (.xlsx), while exports in .jsonl and .parquet formats are also included for machine learning pipelines. The directory structure is as follows.

- 1) *multilingual_datasets/*: This contains the core dataset files organized by domain (grocery/, fashion_retail/, consumer_tech/). Each subfolder contains one .xlsx file per language, with three columns: *item* (English product name), *prompt* (native-language user utterance), and *response* (native-language assistant reply).
- 2) *reproducibility_artifacts/*: These include prompt pattern JSONs, item catalogs, language maps, seed files, Python scripts for generation, and export files (.xlsx, .jsonl, .parquet).
- 3) *evaluation_benchmark_sets/*: This contains curated subsets for few-shot testing, latency benchmarks, and ablation experiments.
- 4) *synthetic_resources/*: These include a primary prompt library for all languages.
- 1) *metadata files*: The metadata files are the README.md, LICENSE, and CITATION.cff for documentation, licensing, and citation [4].

TABLE II. Summary of Dataset Files and Their Contents

File/Directory	Format	Description
multilingual_datasets/<domain>/<language>.xlsx	Excel (.xlsx)	Core dialogue files; each row contains <i>Item</i> , <i>Prompt</i> , <i>Response</i> —one file per language per domain.
reproducibility_artifacts/prompts/*.json	JSON	Prompt pattern libraries with placeholders (e.g., item) for each domain-language pair.
reproducibility_artifacts/items/*.json	JSON	Product name lists for grocery, fashion retail, and consumer tech domains.
reproducibility_artifacts/languages/languages.json	JSON	Language-to-ISO mapping file.
reproducibility_artifacts/seeds/seeds.txt	Text	Fixed random seeds for deterministic dataset generation.
reproducibility_artifacts/scripts/*.py	Python scripts	Main generation script (generate_synthetic_data.py), evaluation and latency scripts, LoRA adapter training utilities.
reproducibility_artifacts/*.jsonl, *.parquet	JSONL/Parquet	Alternative formats of the dataset for direct ML ingestion.
evaluation_benchmark_sets/*.xlsx	Excel (.xlsx)	Benchmark subsets for few-shot, latency, and ablation evaluations.
synthetic_resources/synthetic_prompts_primary_native_transformed_list.xlsx	Excel (.xlsx)	Consolidated primary prompt bank across all languages.
README.md, LICENSE, CITATION.cff	Text	Documentation, license (CC-BY 4.0), and citation metadata.

TABLE III. Schema of Core Dialogue Files (Multilingual_Datasets/<Domain>/<Language>.xlsx)

Feature	Type	Description
Item	String	English product name inserted into the prompt template (grounds the dialogue in a specific retail item).
Prompt	String	User-side utterance in the target (native) language, domain-conditioned, and product-aware.
Response	String	Assistant-side reply in the target (native) language, aligned to the given prompt.

Table II has the details about the main files and their contents.

Each dialogue file follows a consistent three-column schema, as summarized in Table III [9]. An illustrative item–prompt–response example from the published dataset is provided in Table S1 of the supplementary material to clarify the data structure.

INSIGHTS AND NOTES

The Lingu10 dataset serves as a multilingual conversational artificial intelligence (AI) and low-resource language modeling resource, which delivers benefits to various NLP research fields.

Recommended Use Cases

The Lingu10 dataset is best suited for bootstrapping, augmentation, and fine-tuning of multilingual conversational AI models rather than direct standalone training. Its synthetic yet culturally consistent dialogues make it valuable for initializing models, enriching low-resource corpora, and benchmarking cross-lingual adaptation performance.

Caveats and Limitations

The synthetic nature of Lingu10 conversations prevents it from showing the natural imperfections that occur in human communication through typos, slang, and unfinished

statements. The method protects privacy and maintains quality consistency, but it creates difficulties for models trained on this dataset when they encounter unstructured real-world dialogues. The current release has a limited scope, as it only supports three domains: grocery shopping, fashion, and consumer technology. These are important for retail applications, but they do not cover the full range of customer service scenarios. The automatic language detection tool (langdetect) operates effectively in most situations; however, it fails to identify short texts and becomes confused by Chinese and Japanese languages because they share similar writing systems. Future releases may include anonymized real customer dialogues to increase naturalness while maintaining privacy.

Special Case Usages

The developers of Lingu10 built the dataset for chatbot applications, yet the system functions for multiple other applications. The tool enables researchers to conduct evaluation tests across different languages and domains and generates artificial data for both generation and refinement studies [7]. The model offers excellent capabilities for studying style variations and cultural nuances, as well as learning patterns in natural language generation (NLG). The dataset enables researchers to evaluate synthetic dialogues against human-written dialogues, which allows them to determine the advantages and limitations of synthetic corpora for NLP applications.

Potential Extensions

The main advantage of Lingu10 is its simple reproduction and extension capabilities [8]. The system enables users to add new domains through the development of additional prompt libraries and product catalogs for healthcare, travel, and finance sectors. Support for new languages can be introduced in the same way. The framework of Lingu10 functions as an adaptable system that enables researchers

to construct extensive multilingual dialogue resources that expand according to community requirements. Future domain additions (e.g., healthcare, travel, and finance) will be released as independent DOI versions to ensure explicit versioning and reproducibility.

SOURCE CODE AND SCRIPTS

All the source code and reproducibility artifacts for Lingu10 are published openly with the dataset on IEEE DataPort [10]. This package provides everything needed to regenerate or extend the dataset.

The main script for data generation (`generate_synthetic_data.py`) creates balanced prompt–response pairs for all language–domain pairs. The script produces output files in three formats, including `.xlsx`, `.jsonl`, and `.parquet` for easy integration with various workflows. The package also includes prompt and item libraries (`prompts/*.json`, `items/*.json`) that define the dialogue templates and domain-specific product lists used to create native-language conversations.

The `languages.json` file and `seeds.txt` file serve as reproducibility tools because they maintain ISO code stability, ensuring identical results across different runs. The project contains three utility scripts that enhance its functionality: `evaluate_model.py` for evaluation operations, `run_latency_tests.py` for performance benchmarking, and `train_adapter.py` for adapter-based training experiments. A batch script (`batch-commands.txt`) enables users to generate the entire dataset with a single command, utilizing fixed seeds and uniform parameters.

The dataset becomes transparent, reproducible, and simple to modify through the combination of these resources. The researchers can reproduce the original release and verify the reported statistics and extend Lingu10 to new domains and languages by simply updating the prompt libraries and product catalogs.

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The authors contributed equally to this article.

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